

AI-Assisted Air Quality Index Analyzer

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Problem Overview



Problem Statement:

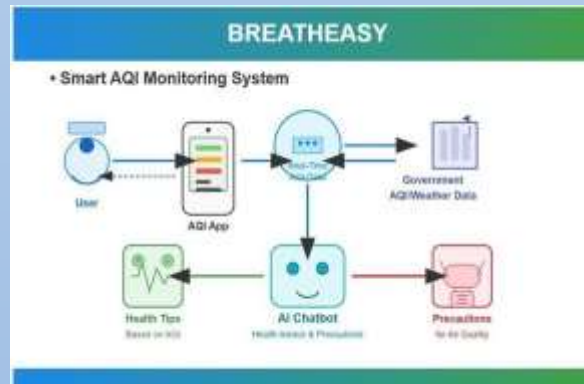
Air pollution is a significant global health risk affecting billions of people worldwide. Current air quality monitoring systems often lack personalization, real-time accessibility, and actionable insights for individuals. There is a critical need for a comprehensive solution that not only reports Air Quality Index (AQI) data but also provides personalized health recommendations based on individual sensitivity and location-specific air quality conditions.

Motivation:

- Rising global air pollution levels significantly impact public health, causing respiratory diseases, cardiovascular problems, and reduced life expectancy
- Need for personalized air quality information that translates complex AQI data into actionable health advice

System Overview

System Architecture



Novelty / Innovation :

- Unlike traditional systems, this project combines real-time AQI data with AI interaction to improve user accessibility, engagement, and guidance.
- AI-powered chat-bot that provides personalized health recommendations based on user queries and current AQI levels.

Impact & Results



- **How it Works**
System pulls real-time AQI data from government sources. Personalized health tips and precautions are delivered through the app.
- **Its Limitations**
Dependence on accuracy and availability of government AQI data sources.
- **Its Future Works**
Mobile App Integration: Develop and integrate the website functionalities into a mobile application for wider accessibility.



QR Code : Link to GitHub



QR Code: Link to Video